

SPECIFICATIONS

For

Compressor Replacement



Date: July 11, 2025

Prepared By:

TOBYHANNA ARMY DEPOT

ENGINEERING DIVISION
11 HAP ARNOLD BOULEVARD
TOBYHANNA, PA 18466-5100

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SCOPE OF WORK
08/15

PART 1 GENERAL

1.1 PROJECT DESCRIPTION

The contractor is responsible to remove and dispose of the existing air compressor and replace it with a new unit that meets specification section 22 15 19.19 20. This includes the decommissioning and removal of the existing compressor system and the procurement, delivery, installation, and commissioning of a new compressor. The new unit must be fully operational and compliant with applicable standards upon installation. The contractor shall conduct all necessary testing and calibration, and shall submit documentation confirming successful commissioning and functional performance. The contractor is responsible for providing all necessary personnel, equipment, supplies, facilities, transportation, tools, materials, and supervision to complete this contract. Any exceptions shall be submitted in writing.

1.2 PERIOD OF PERFORMANCE

Period of performance for this task order shall be 90 days after notice to proceed.

1.3 LOCATION

The work is located at Tobyhanna Army Depot, Tobyhanna, PA. The exact location will be shown by the Contracting Officer's Representative (COR).

1.4 EXISTING WORK

Protection of Existing Vegetation, Structures, Equipment, Utilities, and Improvements:

- a. Remove or alter existing work in such a manner as to prevent injury or damage to any portions of the existing work which remain.
- b. Repair or replace portions of existing work which have been altered during operations to match existing or adjoining work, as approved by the COR. At the completion of operations, existing work must be in a condition equal to or better than that which existed before new work started.

-- End of Section --

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WORK RESTRICTIONS

05/24

PART 1 GENERAL

1.1 CONTRACTOR ACCESS AND USE OF PREMISES

1.1.1 Activity Regulations

Ensure that Contractor personnel employed on the Activity become familiar with and obey Activity regulations including safety, fire, traffic and security regulations. Keep within the limits of the work and avenues of ingress and egress. Wear hard hats in designated areas. Do not enter any restricted areas unless required to do so and until cleared for such entry. Mark Contractor equipment for identification.

1.1.1.1 Escorting Policy

Primary onractors, associated sub-contractors and their employees must be able to obtain and hold unescorted access to the installation. Any contractor personnel needing a government escort will be denied access to the installation for this contract unless otherwise approved in writing by the COR when special circumstances apply.

1.1.1.2 No Smoking Policy

Smoking is prohibited within and outside of all buildings on installation, except in designated smoking areas. This applies to existing buildings, buildings under construction and buildings under renovation. Discarding tobacco materials other than into designated tobacco receptacles is considered littering and is subject to fines. The Contracting Officer's Representative (COR) will identify designated smoking areas.

1.1.2 Working Hours

Regular working hours must consist of an 8 1/2 hour period established by the Contracting Officer's Representative, between Monday through Friday, excluding Government holidays.

1.1.3 Work Outside Regular Hours

Work outside regular working hours requires Contracting Officer's Representative approval. Make application 15 calendar days prior to such work to allow arrangements to be made by the Government for inspecting the work in progress, giving the specific dates, hours, location, type of work to be performed, contract number and project title. Based on the justification provided, the Contracting Officer's Representative may approve work outside regular hours. During periods of darkness, the different parts of the work must be lighted in a manner approved by the Contracting Officer's Representative. Make utility cutovers after normal working hours or on Saturdays, Sundays, and Government holidays unless directed otherwise. TYAD normal working hours are Monday thru Friday 0700-1600.

1.1.4 Occupied Building

The Contractor shall be working in an existing building which is occupied.

Do not enter the building without prior approval of the Contracting Officer's Representative.

The existing buildings and their contents must be kept secure at all times. Provide temporary closures as required to maintain security as directed by the Contracting Officer's Representative.

Provide dust covers or protective enclosures to protect existing work that remains and Government material located in the work area during the construction period.

Relocate movable furniture as required to perform the work, protect the furniture, and replace the furniture in its original locations upon completion of the work. Leave attached equipment in place, and protect it against damage, or temporarily disconnect, relocate, protect, and reinstall it at the completion of the work.

The Government will remove and relocate other Government property in the areas of the building scheduled to receive work.

-- End of Section --

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CONTRACTOR SAFETY STANDARDS FOR TYAD
05/24

PART 1 GENERAL

Below are some "installation specific" safety standards which contractors are required to adhere to while performing any work on Tobyhanna Army Depot. It is the contractor's responsibility to always incorporate these particular requirements into their daily operations while under contract and to institute these specifics into their overall site safety controls. These do not substitute or deflect in any way the requirement to institute and adhere to all OSHA 29 CFR 1910 & 1926 and USACE EM385-1-1, site safety standards as well as all applicable contractual Federal Acquisition Regulations while performing the work under this contract at Tobyhanna Army Depot.

It is the responsibility of the contractor to enforce the below requirements as well as all current revisions of the above referenced standards. Failure to enforce safety requirements on the depot can result in immediate stop works, safety stand down orders and even complete termination of the contract.

1.1 IMPORTANT PHONE NUMBERS

Fire or Emergency: 911
Fire Department - (Non-Emergency):
Security:
Safety Office:

The COR will provide a list of necessary on-site contact numbers to the contractor upon contract award.

1.2 SUBMITTALS

Government approval is required for submittals with a "G" designation; submittals not having a "G" designation are for information only.

SD-06 Test Reports
Monthly Exposure Reports
Notifications and Reports
Accident Reports; G
LHE Inspection Reports

SD-07 Certificates
Activity Hazard Analysis (AHA); G
Confined Space Entry Permit
Hot Work Permit

PART 2 CONTENTS

2.1 THE CONTRACTOR IS SOLELY RESPONSIBLE AND LIABLE FOR JOB- SITE SAFETY

- a. Review of the project plans and other documents by any government appointed official does not constitute an acceptance of Federal responsibility or liability for the adequacy of the safety measures identified for the job or for the Contractor's compliance with all

applicable safety rules and regulations.

- b. The Contractor remains solely responsible and liable for job-site safety at all times during the term of the contract. No outside personnel (other than ET, COR, TYAD Safety, or other government appointed contract personnel) are allowed onto the job site at any time during contract without prior permissions and/or escort by the ET, COR or KO.
- c. Tobyhanna Army depot has restrictions on the use of fossil fuels in all buildings. When it is determined that electric or other alternatively fueled vehicles are insufficient or unavailable for movement of material within the buildings on the installation, fossil-fueled vehicles can be used only when all alternatives of accepted electric material movement equipment have been considered, and the last option is use of a fossil fuel vehicle or project failure. This includes any situations where safety of employees and equipment may be jeopardized. Convenience will never be an acceptable justification to utilize fossil-fueled vehicles. The Government Designated Representative (GDR) will notify the appropriate supervisor(s) or leader(s) in the area where the need to utilize a fossil fuel vehicle is required. The supervisor / leader will take the necessary steps to notify the employees in the work area of the option to leave the area during the material movement process. The GDR and Contractor will also document all actions the Contractor took to utilize electric or other alternatively fueled vehicles. The documentation will specifically state why electric or alternatively fueled vehicles are insufficient and all mitigation efforts used to minimize the effects of fossil-fueled vehicles to include information as to any appropriate scrubbers that can be utilized along with any ventilation or exhaust systems that can be used to externally vented to the outside of the facility. The documentation must be submitted to the TYAD Safety office for approval prior to any equipment being used. A final copy of the Approved documentation will be provided to the TYAD Safety Division.
- d. All direct supervision, including subcontractors, must as a minimum, complete an OSHA 30 hour safety course. These certificates must be submitted as part of the site safety plan BEFORE activities start.
- e. **RED CARD PROGRAM** - Tobyhanna Army Depot utilizes a program to allow employees to stop an action which is considered unsafe. Employees carry small red cards that can be thrown down when employee sees an action they consider unsafe. Work ceases until TYAD management can decide if it is safe or not. Contractors are required to comply with this program. If a TYAD employee presents the red card to
- f. All contractor personnel, to include subcontractors, vendors, suppliers, or visitors, when performing any type of work activity, or gaining access through an occupied area or shop - must have all applicable PPE that is required for the area to include, but not limited to, hardhats, safety glasses, hearing protection, and respirator protection. Notify both COR & Area Supervision, for access approval and prior to any work activities being performed.

2.2 CODE COMPLIANCE

As noted in the Tobyhanna Army Depot Safety Regulation 385-1 - TYAD follows requirements and is certified as an ISO 45001 facility.

Accordingly, the Contractor shall comply and maintain these high safety standards and provide full compliance with all applicable safety regulations and standards. The Contractor shall maintain, monitor, and enforce employee compliance with all applicable rules, regulations, and codes.

Tobyhanna Army Depot requires full contractor compliance with all safety regulations and standards. The Contractor is responsible for maintaining, monitoring and enforcing all rules, regulations and codes, by ALL personnel working for the contractor including all subcontractors. These codes include:

- a. Title 29 Code of Federal Regulations-1910 (OSHA General Industry Standards)
- b. Title 29 Code of Federal Regulations-1926 (Construction Industry Standards)
- c. National Fire Protection Association Codes
- d. Uniform Facilities Code
- e. Unified Facilities Guide Specifications (UFGS)
- f. USACE EM385-1-1
- g. All other codes as required to maintain safety standards.

2.3 JOB HAZARD ANALYSIS (JHA) / ACTIVITY HAZARD ANALYSIS (AHA)

Prior to the start of work, the contractor shall prepare a JHA / AHA for each phase of work that will be done under the contract. This will be completed in accordance with all applicable OSHA safety requirements. A phase is any operations involving a certain type of work. Examples include demolition, excavation, masonry work, concrete pouring, roofing, and electrical wiring. Work done by each subcontractor is also a phase. The JHA/AHA will:

- a. List the activity being performed and identify the sequence of work steps.
- b. List the hazards associated with each step, and the procedures and training required to eliminate or reduce the risk to an acceptable level.

2.4 SITE SPECIFIC SAFETY & ACCIDENT PREVENTION PLANS

Contractors will submit a written Site Specific Safety / Contractor Accident Prevention Plan prior to beginning activities. The plan must follow the guidelines in OSHA 1926:

- a. Who will implement the plan, including who is responsible for safety and accident prevention?
- b. A means for coordinating and controlling subcontractors and suppliers.
- c. Safety training.
- d. Who will investigate accidents?

- e. Emergency response.
- f. Job site cleanup and safe access.
- g. Public safety requirements such as signs and barricades

2.5 FALL PROTECTION

- a. Personnel performing works at a height of six feet or more from any lower level shall comply with OSHA 29 CFR 1926.500 through 1926.503 safety standards.
- b. There are currently no certified anchorage points or permanent guardrail systems on any TYAD roof. Most roofs of the depot are more than 20 feet high.
- c. Contractors and their employees must be protected from falling off the edge of the roof, regardless of how briefly the employees are at the edge of the roof.

2.6 ELECTRICAL SAFETY

Follow the current Tobyhanna Army Depot standard. This standard is based on NFPA 70E.

Copies of the TYAD Electrical standard may be obtained by request from the ET, COR or KO on the project.

2.7 DEPOT REQUIRED PERMITS / APPROVALS

- c. **Confined Space**: All confined space works on TYAD are to be considered as permit required entries, (unless reclassified by the TYAD Safety office,) and Contractor must first request and be provided with an approved "signed" permit from TYAD Safety Office prior to any confined space entry. Contractor will refer to all OSHA requirements and must have a confined space plan and all appropriate safety equipment. Contractor is to request approval from TYAD Safety office through COR prior to entry.
- d. **Burning / fire**: Hot work permits must be obtained daily from the Fire Department.

For sealed sources of radiation, i.e. Moisture Soil Density Gauges, the application must include a valid Nuclear Regulatory Commission (NRC) Radioactive Materials License that allows the applicant to use the source as specified in the ARP application, current leak test certificates, operator training records and calibration certificates for any equipment that may be utilized at TYAD.

For machine produced ionizing radiation sources, i.e. X-ray Equipment, the application must include the appropriate state authorization that allows the applicant to use the source as specified in the ARP application along with operator training records and calibration certificates for any equipment that may be utilized at TYAD.

ARPs will not be issued for more than 12 months at a time.

2.8 CONFINED SPACES

- a. All Confined Spaces at TYAD are considered as Permit-Required Confined Spaces unless reclassified by TYAD Safety office after a thorough review of the area and the work being planned.
- b. Contractors must first notify the COR and receive TYAD Safety office approvals prior to gaining access or opening any confined space area.
- c. Contractors must fully comply with all OSHA regulations for Confined Space Operations, including on-site use and availability of a Confined Space Entry Permit / Checklists, along with prior approvals of their Accident Prevention Plan and associated Activity Hazard Analysis (AHAs) for the work to be performed.
- d. The contractor and/or the COR will then notify the TYAD Safety office and TYAD Fire and Emergency Services of the Confined Space Entry timelines and schedules.
- e. Upon completion, the contractor will notify the TYAD Safety Office and discuss whether any new hazards were added to the Confined Space and will submit the completed Confined Space Entry Permit / Checklist for record.

2.9 LOCKOUT/TAGOUT PROCEDURE

- a. The Contractor shall perform this work in accordance with 29 CFR § 910.147-The Control of Hazardous Energy requirements.
- b. The Contractor shall notify the COR that a lockout/tag out system is going to be used.
- c. The contractor is to request the COR to notify them if TYAD is using any lockouts / tag outs in the area.
- d. The Contractor shall locate and identify all isolating devices.
- e. The Contractor shall shut down the equipment normally.
- f. The Contractor shall isolate all energy using appropriate methodology.
- g. After all work is complete, the Contractor shall remove the energy isolating device and restore the equipment to service.
- h. The Contractor shall notify the COR that energy is again restored.

2.10 PROTECTION TO THE DRINKING WATER SYSTEM

The contractor will not perform any work on the depot potable water system until prior approval is obtained from the Environmental Branch and the certified operator through the Directorate of Installation Services COR.

2.11 PROTECTION OF PERSONNEL

At no time is any depot employee to be put at risk to illness or injury. No actions are exempt from this requirement.

- a. Where pedestrian and driver safety is endangered, use appropriate traffic barricades. Use anchor barricades to prevent displacement by

wind. Institute PA DOT regulations and OSHA requirements. Notify the COR prior to beginning such work.

- b. Where demolition is required, a Demolition Plan describing the steps and individual Definable Features of Works (DFOWs) as well as safety protocols, controls and oversight is required to be submitted to the COR for review prior to works being performed. Continuously evaluate the condition of the structure being demolished and take action to protect personnel working around the project site. No structural element will be left standing without sufficient support to prevent collapse.
- c. Contractor shall provide personal protective equipment (PPE) and provide training for use, for all visitors requiring entry into the regulated area.
- d. Subcontractors, vendors, and delivery personnel are the responsibility of the prime contract and must follow the same regulations the prime contractor does.

2.12 FIRE AND EMERGENCY SERVICES

Hot Work Permits will be issued by Fire Department upon request. All requests must be made at least thirty (30) minutes prior to the expected start time by contracting the fire department.

- a. Contractors are required to have fully charged and operable contractor provided fire extinguisher, in quantities and properly spaced per OSHA requirements that are appropriate for the type of possible fire. Fire extinguisher shall be immediately available for inspection by the fire department upon an inspection of the worksite and shall remain at the work site at all times. At a minimum, 10 pound class 4A, 60B or C multi-purpose dry chemical fire extinguisher or equivalent shall be provided.
- b. A fire watch must be in place during the hot work and for 30 minutes after the HWP operation. Upon completion of the thirty (30) minute, post hot work fire watch, the contractor shall contact the fire department to report the work is complete.
- c. Use of a plasma torch will require special approval by the fire department.
- d. Any fire, even one that is extinguished, must be reported to the fire department.
- e. Ambulance services are provided at a cost to contractors. The contracted ambulance service provider will transport to the nearest hospital. Advanced Life Support, helicopter transport, etc. is all at the contractor's expense.
- f. The fire department will respond to all fire, ambulance, hazmat and confined space incidents.

2.13 IF AN INCIDENT HAPPENS

- a. **Any serious injury or medical emergency:**
 - Immediately call emergency responders.
 - Dial 911 and give accurate location and nature of emergency. Make

- sure to tell the operator that you are calling from Tobyhanna Army Depot.
- Send someone to the nearest exit to assist responders.
 - Treat victims to the best of your ability.
 - Notify COR / call work order desk if there is any blood or body fluids that must be cleaned up.
- b. **In the event of the smell of natural gas:**
- Call dial 911 and give accurate location and nature of emergency. Make sure to tell the operator that you are calling from Tobyhanna Army Depot.
 - Have all personnel evacuate the area if the smell is significant.
 - Upon arrival of fire personnel, provide information as to specific equipment and exact location of potential leak.
- c. **In the case of a fire:**
- If the fire is small, use a fire extinguisher and call 911.
 - Otherwise, dial 911 and pull the fire alarm.
 - Evacuate the building.

Accident/Injury Reports and OSHA 300 Reports:

The contractor shall report to the COR and KO, - Immediately but no later than eight (8) hours, any Fatality, In-patient hospitalization, Amputation, Eye loss or Property damage over \$600,000. Similarly, contractor shall report to the COR and KO, - Immediately, but no later than twenty-four (24) hours, all other accidents or near misses.

Notifications shall be followed up within five (5) working days of all incidents (reportable or not) with an OSHA 301 report including all available facts relating to each instance of personal injuries and of loss / damage to Government property.

For any accident that results in illness or injury that requires more than first aid treatment, TYAD Safety & Health Office has the option to have the contractor immediately secure the accident area until released by the accident investigation authority as designated by the Commander. If the Government elects to conduct an investigation of the accident, the contractor shall cooperate fully and assist authorities until the investigation is complete. If requested after the investigation is completed, the contractor shall provide a Corrective Action Plan related to the incident to the COR & KO.

COR to provide copies of all OSHA 301 incident reports and any Correction Action Plans to the TYAD Safety office when available.

2.14 APPLICATION OF ANY ODOR PRODUCING MATERIAL

- a. Any application of any paint, epoxy, adhesive or anything else that produces an offensive odor must be coordinated with the COR or POC before starting. This is particularly true of work in a building occupied by Tobyhanna employees.
- b. It is highly recommend that this type of work be planned for off hours, weekends, or holidays to avoid shutdowns of operations due to these offensive odors. Odors during curing times for products shall also be considered.

2.15 CONTROL OF RESPIRABLE CRYSTALLINE SILICA

- a. Contractors are required to follow the OSHA standard 29 CFR 1926.1153 which requires employers to limit worker exposures to respirable crystalline silica and to take steps to protect their workers and TYAD personnel.
- b. Contractors shall establish and implement a written exposure control plan that identifies tasks that involve exposure and methods used to protect workers, including procedures to restrict access to work areas where high exposures may occur.
- c. Restrict housekeeping practices that expose workers to silica by utilizing engineering controls such as local exhaust ventilation, using "wet spray" methods when cutting or drilling and providing proper respiratory protection and other PPE as applicable.

IN ANY CONFLICT BETWEEN STANDARDS, THE STRICTER OF THE TWO SHALL APPLY. IF UNSURE CONTACT THE COR / ET VIA PROPER RFI PROTOCOLS FOR VERIFICATION THROUGH THE TYAD SAFETY OFFICE.

-- End of Section --

SECTION 01 35 53.00 99

TYAD SECURITY OPERATIONS REQUIREMENTS
9-25

PART 1 GENERAL

1.1 SUBMITTALS

Government approval is required for submittals with a "G" designation; submittals not having a "G" designation are for information only. Submit the following in accordance with Section 01 33 00.00 99 SUBMITTAL PROCEDURES:

SD-01 Submittals

- OPSEC Standing Operating Procedure (SOP)/Plan; G
- Initial List of Verified/Eligible Candidates; G

SD-07 Certificates

- AT Level I Training
- Level I OPSEC Training
- OPSEC Awareness Training
- Threat Awareness And Reporting Program (TARP) Training

1.2 FORCE PROTECTION CONDITION (FPCON)

Should the Force Protection Condition (FPCON) at the installation change, the Government may require changes in contractor security matters and/or processes.

If based on the requirements of this contract, contractor employees may be designated as mission-essential. Mission-essential contractor employees are required to report for duty and remain on duty during declared emergencies and/or elevated FPCON levels unless otherwise directed by the Contracting Officer (KO) via the appropriate Contracting Officer's Representative (COR).

Unscheduled gate closures by Law Enforcement may occur at any time causing all personnel entering or exiting a closed installation to experience a delay. This cannot be predicted or prevented. Contractors are not compensated for unexpected closures or delays.

Vehicles operated by contractor personnel are subject to search pursuant to applicable regulations. Any moving violation of any applicable motor vehicle regulation may result in the termination of the contractor employee's installation driving privileges.

The contractor's employees shall become familiar with and obey the regulations of the installation, including law enforcement, fire, traffic, safety and security (protecting National security) regulations while on the installation. Contractor employees should only enter restricted areas when required to do so and only upon prior approval. All contractor employees shall carry proper identification with them at all times. The contractor shall ensure compliance with all regulations and orders of the installation which may affect performance.

1.3 EMERGENCY COMMUNICATIONS AND SERVICES

If assigned to TYAD for seven-days or longer, contractors and subcontractor employees will enroll in the ALERT! Mass Warning Notification System (MWNS). CAC is required to register for ALERT!. If a CAC has not been deemed necessary for the contractor or subcontractor, the COR will provide the necessary information to TYADs Antiterrorism Officer or TYADs Emergency Manager in the Security Operations Branch for manual input into the ALERT! System.

Alert! MWNS is NOT optional - it is a Headquarters Department of the Army (HQDA) requirement to be a registered user.

Due to the life-safety implications of the information being relayed and the requirement to provide immediate alerts and warnings, personnel assigned to TYAD must ensure that their personal contact information, including after-duty hours contact information, as appropriate (e.g., personal cellular phone numbers or landline phone numbers), e-mail addresses, home address, etc., are entered into the system and regularly updated or verified every 90 days to remain current and accurate.

In the event of a life-threatening emergency or natural disaster, COR will account for assigned contractors. COR will send personnel accountability report to Emergency Operations Center (EOC).

Base closure announcements will be disseminated via ALERT! MWNS.

1.4 ANTITERRORISM (AT) LEVEL I TRAINING

In accordance with AR 525-13, all contractor and subcontractor employees requiring access to Army installations, facilities, or controlled access areas shall complete AT Level I awareness training annually. This training must be completed prior to being granted installation access and beginning work. The contractor is responsible for ensuring this requirement is met and for providing certificates of completion for each employee to the COR upon request. The web-based training is available at <https://jko.jten.mil> (Course: JS-US007).

AT Level I and iWATCH training is not required to be in person. While in-person training is an option, the most common and widely accepted method is web-based. Army regulations explicitly provide multiple ways to meet the training requirement to ensure flexibility for all personnel, regardless of their location or access to a classroom setting.

1.5 iWATCH ARMY TRAINING

This training will be used to inform employees of the types of behavior to watch for and instruct employees to report suspicious activity to the COR. This training shall be completed within 30 calendar days of contract award and within 30 calendar days of new employees commencing performance with the results reported to the COR NLT 30 calendar days after contract award. Contractors and all associated sub-contractors will be briefed by the TYAD Antiterrorism/Force Protection officer. It is recommended this training be conducted as part of an initial entry briefing or unit/organization newcomer's briefings. Send training requests to TYADs Antiterrorism Officer. Contractor employee training will be documented on a local form.

1.6 ACCESS AND GENERAL PROTECTION/SECURITY POLICY AND PROCEDURES

Contractor and all associated sub-contractor employees shall comply with

applicable installation, facility and area commander installation/facility access and local security policies and procedures (provided by government representative). The contractor shall also provide all information required for background checks to meet installation access requirements to be accomplished by TYAD Law Enforcement. Contractor workforce must comply with all personal identity verification requirements as directed by DoD, HQDA, and/or local policy.

Contractor and all associated sub-contractor employees shall comply with adjudication standards and procedures using the National Crime Information Center Interstate Identification Index (NCIC-III) and Terrorist Screening Database (TSDB) (Army Directive 2014-05/AR 190-13), applicable installation, facility and area commander installation/facility access and local security policies and procedures (provided by the government representative), or, at OCONUS locations, in accordance with status of forces agreements and other theater regulations.

A background check and approval from TYAD Law Enforcement is required for all contractor and subcontractor personnel prior to on-site access at TYAD. All persons seeking entrance to TYAD shall submit to and comply with all security standards and requirements in force at the time such persons are seeking entry. All contractors, regardless of resident status or citizenship, will be subject to vehicle search and intense in-processing by TYAD security personnel prior to being granted access to TYAD. This security screening process may be time consuming, and access may be delayed or denied. The contractor shall ensure the required government access request form is completed. COR will provide specifics to the contractor after award and at least ten days prior to the expected visit date for completion. All the required fields on the form shall be accurate and complete. This requirement is inclusive of on-site supervisory or managerial personnel and sub-contractor personnel that the contractor anticipates will be performing work or visiting on-site. This security screening does not relieve the contractor of any responsibilities to conduct thorough pre-employment background checks and drug screening. Contractor workers will not be granted access to the work site until security screening is completed and access is approved. Any contractor personnel on-site who fail screening will not be permitted further access to TYAD.

Submit the completed local form(s) to the COR. Ensure contracts include the provisions that check for the possibility of and prevent undocumented workers for inclusion in contracted work related to Army missions.

The company will ensure that its employees entering Army-controlled installations or facilities have obtained access badges and passes in accordance with facility regulations and that these badges and passes are obtained in advance so as not to delay the accomplishment of contracted services.

1.7 PRE-BID SITE VISIT CONTROLS

Any pre-bid site visits held before the contract award will be conducted under strict security controls. These protocols-including mandatory pre-registration for baseline vetting and constant escorting-apply to all potential bidders, regardless of any previous work history with TYAD or current installation access. During this public solicitation phase, a contractor's prior access or familiarity with the installation does not grant them a "need-to-know" for a new requirement. All bidders will be treated as external entities, and the site visit is strictly for a visual

inspection of physical conditions only.

1.8 PRE-AWARD INFORMATION RELEASE

Conversely, all project specifics-including exact locations, detailed plans, and mission context-can and must be provided to the contractor only after the contract is awarded and their personnel have completed all mandatory security training (AT, OPSEC, CUI). At this stage, the contractor has an established "lawful government purpose," and the controlled release of detailed information is appropriate and necessary for the performance of the contract.

1.9 COLLECTION OF BADGES

The Contractor shall account for all forms of Government-provided identification issued to the Contractor employees in connection with performance under this contract. The Contractor shall return such identification to the issuing agency at the earliest of any of the following, unless otherwise determined by the Government:

- a. When no longer needed for contract performance.
- b. Upon completion of the Contractor employee's employment.
- c. Upon contract completion or termination.

The Contracting Officer may delay final payment under a contract if the Contractor fails to comply with these requirements. The Contractor shall insert the substance of this clause, including this paragraph, in all subcontracts when the subcontractor's employees are required to have routine physical access to a Federally-controlled facility and/or routine access to a Federally-controlled information system. It shall be the responsibility of the prime Contractor to return such identification to the issuing agency in accordance with the terms set forth in of this section, unless otherwise approved in writing by the Contracting Officer. The company will return all issued U.S. Government Common Access Cards, installation badges, and/or access passes to the COR when the contract is completed or when a contractor employee no longer requires access to the installation or facility. Contractor personnel will obtain a vehicle pass for access to the military installation and Common Access Cards (CAC) for computer access, if applicable.

1.10 INSIDER THREAT PROGRAM

The contractor will establish and maintain an insider threat program to gather, integrate, and report relevant and available information indicative of a potential or actual insider threat, consistent with E.O. 13587 and Presidential Memorandum "National Insider Threat Policy and Minimum Standards for Executive Branch Insider Threat Programs."

1.11 OPERATIONS SECURITY (OPSEC)

OPSEC is not a one-time checklist, it is a dynamic, ongoing process embedded in every phase of planning, execution, and daily operations. It requires continuous critical thinking and vigilance from everyone who interacts with government information or operates on a government installation. This includes military personnel, Department of Defense civilians, contractors, subcontractors, and family members. OPSEC is not just a policy, it's a mindset. Its success depends on consistent awareness and execution by every member of the TYAD community.

At TYAD, all observations, communications, and experiences must be

protected and must not be shared with anyone lacking a lawful government purpose or a validated business need-to-know. This disciplined approach ensures that emerging threats, evolving vulnerabilities, and shifting mission parameters are constantly reassessed and addressed to safeguard critical information.

OPSEC applies across the full spectrum of DoD missions, functions, programs, and activities under this contract. Practicing OPSEC daily is a requirement that is essential to mission success and the protection of national security assets.

Failure to properly safeguard sensitive or classified information-whether through negligence, willful misconduct, or ignorance-may result in administrative, disciplinary, contractual, or criminal consequences.

1.12 APPLICABILITY AND MANAGEMENT OF THE CONTRACOR OPSEC PLAN

The requirement for an Operations Security (OPSEC) Plan or Standing Operating Procedure (SOP) is based on the contractor's access to and handling of Critical Information and/or Controlled Unclassified Information (CUI), regardless of where that information resides. It is essential to distinguish between Information Security (INFOSEC), which protects data "at rest" within a system, and OPSEC, which protects data "in use" by governing the procedures and actions of personnel.

When contractor personnel access or handle CUI-such as building floor plans, technical manuals, or operational schedules-they are required to have an approved OPSEC Plan/SOP. This plan is the formal, documented agreement that outlines how the contractor will protect the government's sensitive information. Even when CUI is handled exclusively on Government Furnished Equipment (GFE), the contractor remains responsible for the "human factor." The OPSEC Plan transfers the responsibility for protecting this information from the Government to the contractor by legally obligating them to establish internal procedures for everything from secure physical storage and controlled discussions to the proper destruction of materials. Without this plan, the Government retains all risk for how a contractor safeguards its data.

The management and approval of the contractor's OPSEC Plan is a shared responsibility, with defined roles at each stage. Per government acquisition and security policy, it is the Contracting Officer's Representative (COR) who is ultimately responsible for the initial identification and final enforcement of this requirement. The process is as follows: The COR (designated technical representative) identifies the need for an OPSEC Plan based on the work described and writes the requirement into the Statement of Work (SOW). After contract award, the COR enforces this by ensuring the contractor submits the plan on time. The Contracting Officer (KO) ensures the requirement is legally binding in the final contract and is the sole authority to take contractual action for non-compliance. Finally, the Government OPSEC Officer, as the technical expert, is responsible for reviewing the submitted plan against AR 530-1 and providing the formal approval or rejection back to the COR.

1.13 OPERATIONS TRAINING REQUIREMENTS

In accordance with AR 530-1 (4-2-a), all contractors and subcontractors shall complete OPSEC Level I certification training (Initial and Continual - detailed below).

The Contracting Officer's Representative (COR) serves as the oversight authority: the COR verifies contractual compliance by collecting and reviewing certificates, ensuring they're submitted within the required timeframe - first 30 days of assignment.

According to AR 530-1 (para 4-2) and DoDM 5205.02, there is no "grace period" or minimum number of days on-site before OPSEC Level I Initial and Continual training becomes mandatory-any contractor, subcontractor, or support?service provider who requires access to TYAD information, systems, or facilities must complete both parts of their OPSEC Level I training.

1. OPSEC Level I Initial training

- o To be completed within 30 days of assignment to TYAD and annually thereafter - preferably before the job starts
- o Provides specific information about TYADs OPSEC program
- o Contractors and subcontractors must set up an appointment with TYADs OPSEC Officer for Initial training-COR to provide contact information
- o Conducted by TYAD's OPSEC officer or alternate - training can be in person or via MS Teams (30 minutes) and is tracked by a TYADs OPSEC Officer via a local training form

2. OPSEC Level I Continual training:

- o Commonly referred to as annual OPSEC training
- o To be completed within 30 days of assignment to TYAD and annually thereafter - preferably before the job starts
- o Provides general information about OPSEC and reemphasizes sound OPSEC practices throughout the contract period
- o Title of the training is: "OPSEC Awareness for Military Members, DoD Employees and Contractors" and is available online - <https://securityawareness.dcsa.mil/opsec/index.htm>
- o Certificates of completion must go to the COR

1.14 OPSEC MEASURES TO PROTECT CRITICAL INFORMATION

All contractors and subcontractors shall implement the following mandatory OPSEC measures during the entire period of performance to protect critical and sensitive unclassified information:

1. Training and Compliance: Complete all mandatory OPSEC training as specified in this contract and actively apply all TYAD OPSEC measures in daily, contract-related operations.
2. Secure Communications: Avoid discussing government operations in public spaces or over unprotected communications (e.g., personal cell phones). Transmit Controlled Unclassified Information (CUI) only through authorized, encrypted channels as directed in the SOW/PWS = DoD Safe Access File Exchange (SAFE) is the accredited method for sending files that contain CUI, Personally Identifiable Information (PII), or other sensitive data that should not be transmitted over

unencrypted, commercial email channels. The files are encrypted during the transfer process.

3. **Public Release and Social Media Prohibition:** Do not disclose or post any images, data, or operational details (e.g., maps, pictures, schedules, equipment, tactics) to any unofficial media, including personal or corporate social media platforms. All information or materials intended for public release require a documented OPSEC review by a TYAD certified OPSEC Level II reviewer. Coordinate all press releases through the COR and TYADs Public Affairs Office.
4. **Reporting Requirements:** Immediately report any attempts by unauthorized third parties to solicit sensitive information to the TYAD Security Operations Branch, the contract POC, and the contractor's Facility Security Officer (FSO).
5. **Credential Control:** Conceal or remove all government-issued badges and identification when off the installation. Do not duplicate, loan, or leave credentials unattended (CACs, badges, and/or parking passes or placards). Report any lost or stolen passes/badges/credentials to TYAD Law Enforcement Branch. The COR will provide all POC contact information.
6. **Protection of Sensitive Data:** Mark and protect internal company data related to this contract (e.g., schedules, deliverables, vulnerability reports) as CUI. All government-provided information, such as drawings and technical data, must be protected in accordance with DoD, Army guidance and guidance within contract package documents.
7. **Information Disposal:** At contract completion or upon Government request, return all government information to the COR or destroy it in accordance with approved TYAD procedures for handling CUI.

1.15 PHOTOGRAPHY OR VIDEOGRAPHY

Contractors and subcontractors who will require photography or videography at TYAD must notify their COR at least fourteen business days in advance and complete OPSEC Level I Initial training. The COR will provide contractors with the contact information for points of contact within the security branch after award. All individuals must follow TYAD policy regarding photography and the use of personal electronic devices. All photos and videos must then undergo an OPSEC Review-per AR 530-1 Chapter 4-to evaluate them against TYAD's Critical Information List and identify any vulnerabilities. That review must be fully documented and signed off by a TYAD employee who holds OPSEC Level II certification (example: TYAD OPSEC Officer or Public Affairs personnel) before any imagery is distributed.

1.16 COMMUNICATIONS SECURITY (COMSEC)

All telephonic and electronic communications with DoD organizations are subject to Communications Security (COMSEC) review and monitoring. The contractor is advised that any calls placed to or received from a DoD organization may be subject to monitoring and recording. The contractor shall ensure frequent dissemination of this notice to all employees who communicate with DoD personnel.

1.17 INFORMATION SECURITY (INFOSEC)

The contractor shall comply with all applicable regulations for the protection of government information. Information handled under this contract is categorized as either Controlled Unclassified Information (CUI) or Classified National Security Information (CNSI).

1.18 CONTROLLED UNCLASSIFIED INFORMATION (CUI)

The handling, marking, safeguarding, and destruction of all CUI shall be in strict accordance with DoD Instruction 5200.48, Controlled Unclassified Information (CUI). Specific contractor responsibilities are detailed in the "Safeguarding CUI" section of this document.

1.18.1 CUI TRAINING REQUIREMENT

Any contractor or subcontractor personnel who will have access to CUI under this contract must complete mandatory CUI training prior to accessing such information. The "DoD Mandatory Controlled Unclassified Information (CUI) Training" course available on the DCSA CDSE Security Awareness Hub (<https://securityawareness.dcsa.mil/cui/>) satisfies this requirement. Completion certificates shall be provided to the COR upon request.

1.18.2 CONTRACTOR RESPONSIBILITY FOR HANDLING CUI

The contractor shall comply with all CUI requirements outlined in Executive Order 13556 and 32 CFR Part 2002. The contractor's responsibilities include, but are not limited to:

Marking: Applying uniform CUI markings to all CUI data and deliverables.

Access Controls: Restricting CUI access to personnel with a lawful government purpose and validated need-to-know

Flow Down: Ensuring all CUI protection requirements are passed down to subcontractors.

Physical Safeguarding: Implementing appropriate physical controls (e.g., locked storage, controlled-access areas) for all CUI, whether in hard copy or on electronic media

Incident Reporting: Reporting any actual or suspected loss or compromise of CUI to the COR immediately

Disposition: Returning all CUI to TYAD control or destroying it in accordance with DoD and Army guidelines upon contract completion

1.19 CLASSIFIED NATIONAL SECURITY INFORMATION (CNSI)

If this contract requires access to CNSI, all such information shall be handled, stored, and protected in strict accordance with AR 380-5, Army Information Security Program, the National Industrial Security Program Operating Manual (NISPOM), and the security classification guidance provided in the DD Form 254 issued for this contract.

1.20 PHYSICAL SECURITY

1.20.1 SAFEGUARDING OF GOVERNMENT PROPERTY

The Contractor shall be responsible for safeguarding all government equipment, information, and property provided for Contractor use. At the close of each work period, government facilities, equipment, and materials shall be secured.

1.20.2 Key Control

The Contractor shall establish and implement methods of making sure all keys and key cards issued to the Contractor by the Government are not lost, misplaced, or used by unauthorized personnel.

No keys issued by the Government shall be duplicated.

The Contractor shall develop a key control procedure, to be included in their Quality Control Plan, which includes procedures for the turn-in of keys by personnel who no longer require access.

The Contractor shall immediately report any lost or duplicated keys to the Contracting Officer (KO). In the event a key is lost or duplicated, the Government may, at its option, re-key or replace the affected locks, and the total cost shall be deducted from the monthly payment due to the Contractor.

-- End of Section --

SECTION 01 57 19.00 06

STANDARD SPECIFICATIONS FOR PROJECTS UNDER THE
NATIONAL ENVIRONMENTAL POLICY ACT
IN ACCORDANCE WITH 32 CFR 651
PROPONENT ELTY-ISP-N
06/25

PART 1 GENERAL

Spill Response Procedures 1-2-3 1. Stop work 2. Call 911 3. Evacuate Area *****Damaged Asbestos?***** Contact Environmental Branch (EB) *****Environmental Questions?***** Contact Environmental Branch (EB)

1.1 AIR POLLUTION CONTROL

The contractor must control fugitive emissions, including dust, during the course of their contract. The contractor must obtain approval from the Environmental Branch (EB) prior to exhausting equipment to the outside. The contractor must not allow any pollutant or particulate matter to be released to the atmosphere at levels that are visible from outside of Tobyhanna Army Depot (TYAD). The contractor must not perform work that will release pollutants or particulate matter to the atmosphere when the wind speed exceeds ten miles per hour and will result in adverse effects to the surrounding areas. TYAD will monitor the wind speed; it is the contractor's responsibility to obtain wind speed information through the Contracting Officer Representative (COR).

1.2 ASBESTOS

The contractor must ensure that all materials used in the performance of this contract are asbestos-free.

Unless specified in the contract, the contractor must not disturb any existing Asbestos-Containing Material (ACM) in the performance of this contract. If ACM, or suspect ACM might be disturbed in performance of this contract, the contractor must avoid coming in contact with the material and immediately notify the COR and the Contract Administrator in writing. The COR must coordinate with EB to have the material tested to determine if there is ACM. If the material is determined to be ACM and the contractor cannot avoid disturbing the material, the COR will notify the Contracting Officer. The Contracting Officer will direct a change pursuant to the contract clauses entitled "Changes" and "Differing Site Conditions." If ACM, or suspect ACM has been disturbed, the contractor must immediately notify the COR and call the EB. The contractor will shut

down and not move any equipment or supplies near the damaged ACM. The contractor will evacuate all non-contaminated contractor personnel from the immediate vicinity. Any contractor personnel thought to be contaminated with asbestos must remain in the area until the EB responds. If the damaged material is determined to be ACM and there is potential for further damage, the Contracting Officer will direct a change pursuant to the clauses of the contract clauses entitled "Changes" and "Differing Site Conditions."

Asbestos abatement required under the contract as originally awarded must be in accordance with United Facilities Guide Specification 02 83 14 00 10 and as otherwise required in the contract. Asbestos abatement not required under the contract as originally awarded, can be incorporated into the existing contract via contract modification, or by the Government taking responsibility for the asbestos abatement. The method of acquiring the abatement is at the discretion of the Government. The contractor must have an asbestos abatement work plan that has been approved by EB prior to beginning any asbestos abatement work.

1.3 BACKFLOW

The contractor shall have a backflow prevention device installed on all contractor equipment that is connected to TYAD's water distribution system. The contractor shall have a water meter installed to monitor water consumption during all phases of the contract. Water use will be reported monthly to the COR.

1.4 BURNING

The contractor must not burn refuse and debris anywhere on TYAD.

1.5 DRINKING WATER

The contractor must not perform any work on the TYAD potable water system prior to obtaining approval from the EB and the Installation Planning and Maintenance Division certified operator through the COR. If a permit is required due to installation or proposed chemical feed changes, it will be the responsibility of the contractor to obtain all permits associated with the project. Permit applications will first be reviewed and approved by the Environmental Branch before being submitted to regulatory authorities. Any piping or additions added to the TYAD water system must be disinfected following "American Water Works Association circular C651-14 "Disinfecting Water Mains." Installation will be kept outside of drinking water wellhead protection zones when feasible. All new buildings and remodeled buildings will be equipped with a water meter that can easily be read from the exterior of the building.

1.6 ENDANGERED SPECIES

The contractor is responsible for meeting requirements of the Endangered Species Act of 1973. The contractor must not disturb any endangered species, their habitat or offspring during the implementation of this contract.

1.7 EMERGENCY/SPILLS

All emergencies and spills must be reported to the TYAD Fire Department by calling 911 from a TYAD phone or (570) 615-7300. If a 911 call is placed on a cell phone, the call will go to the Monroe County Emergency Office.

Notify the Monroe County Emergency Office that you are at TYAD and the call will be forwarded to TYADs Fire Department. The contractor must ensure all personnel working on site are trained in the proper procedure according to 29 CFR 1910.120 (if applicable) to initiate a spill response to handle the hazardous substances they are working with. The contractor will take the necessary actions to prevent and contain spills of hazardous materials.

1.8 ENVIRONMENTAL AUTOMATION AND CONTROL SYSTEMS

The contractor will not alter, modify, remove or tamper with any environmental automation or control system unless previous arrangement have been made with EB. This includes sensors, programmable logic controllers, equipment housing, power supplies, meters or other hardware/sensor technology.

1.9 HAZARDOUS MATERIALS STORED AND LABELED

The contractor must ensure all hazardous materials (HM) at the work site are properly stored and labeled. The contractor must not leave any HM behind at the completion of the job for any reason. HM must not be stored outside without adequate secondary containment and shelter.

1.10 HAZARDOUS WASTE

The contractor must ensure that all hazardous wastes (HW) at the work site are properly stored and labeled in a pre-approved location designated by the EB. The contractor must provide copies of any shipping documents for HW/Universal Waste/Toxic Substances Control Act waste. If the TYAD EPA number is being used for shipping purposes, only EB is authorized to sign a manifest and the manifest (or copy) must be supplied prior to the day of shipment for review. The EB will keep originals. If HW or waste requiring special handling (e.g., asbestos) is being turned over to the government for disposal, the contractor must notify the EB through the COR when the waste is ready to be moved.

1.11 ISO 14001

TYAD is an ISO 14001 certified facility. All contractor and subcontractor employees on site must comply with TYAD Regulation 200-5, "Environmental Management System (EMS)." The contractor may obtain these regulations through the project COR.

1.12 SAFETY DATA SHEETS (SDSS)

The contractor must submit SDSs for all hazardous materials proposed for use, including paints, solvents, adhesives, etc., to the EB through the COR five working days prior to material being brought on post. As of June 1, 2015, all Material Safety Data Sheets (MSDS) must be replaced with new Safety Data Sheets (SDS). EB will no longer be accepting older legacy MSDS sheets and the contractor must submit the current SDS for the hazardous material. The contractor must keep a copy of all SDSs required for the project at the jobsite.

1.13 NATIONAL POLLUTANT DISCHARGE ELIMINATION SYSTEM (NPDES) PERMITS

The contractor must not perform any work on existing NPDES structures or treatment units unless previously approved in writing by the Installation Planning and Maintenance Division certified operators and the

Environmental Branch. This includes work within the sewage treatment plant, sewage lift stations, sewage conveyance pipes, Industrial Operations Facility pretreatment plant, and storm water sewer systems. If a permit is required, it will be the responsibility of the contractor to obtain all permits prior to work being performed. All permit application packages must first be reviewed by the Environmental Branch prior to being sent to any regulatory authority. All liquids (chemical, cleaner, and lubricants) in containers, drums, and IBC totes are to be stored on secondary containment at all times. The container shall be labeled with the contents inside and an SDS sheet must be provided to EB.

1.14 NOISE

The contractor must not allow the noise level to exceed 65 decibels at any point outside TYAD property. If noise levels exceed 65 decibels, a plan must be prepared by the contractor to mitigate the noise levels and submit to the EB for approval through the COR. The contractor will monitor the fence line to confirm this limit.

1.15 OZONE DEPLETING SUBSTANCES (ODS)

The contractor must be responsible for ensuring that all personnel who perform maintenance and repair activities on refrigeration equipment have been trained and certified by an EPA-approved Section 608 program. The contractor must not use Class I or Class II ODS or install equipment that contains Class I or II ODS. Beginning 1 Jan 26, the contractor must adhere to all AIM Act regulations regarding Hydrofluorocarbon refrigerants, Title 40, Chapter I, Subchapter C, Part 84.

1.16 PAINTS

The contractor must not use paints containing zinc chromate or strontium chromate pigments and paints must not contain a concentration of lead greater than 0.009 percent. Coatings used within permitted paint spray booths must not have a Volatile Organic Compound (VOC) content greater than 3.50 pounds per gallon.

1.17 PEST MANAGEMENT

At no time during the execution of this contract must the contractor provide a food source or harborage for any pests. The contractor must coordinate through the COR to the EB prior to the application of any pesticide (PA state license is required for pesticide applications). All pesticides are required to be approved by the EB. The contractor must report all usage of pesticides the day they are applied through the COR to the EB. After completion of the contract, the contractor must ensure there is no passage for pests to enter facilities or structures related to work performed by the contractor.

1.18 REFUSE AND CONSTRUCTION DEMOLITION DEBRIS REMOVAL AND DISPOSAL

During the performance of all construction, renovation and demolition projects, a minimum of 60 percent of all non-hazardous construction demolition debris shall be diverted from the landfill for reuse or recycling. The contractor shall provide written certification to the COR of the type and tonnage of debris reused or recycled from the contract. All refuse and the construction demolition debris not recovered for reuse or recycling shall be disposed of at a PADEP permitted and Monroe County Municipal Waste Management Authority authorized facility. The contractor

shall comply with the Monroe County Solid Waste Management Plan and PA Act 90, including all licensing requirements. Monroe County has specific landfills that is must be transported to per Monroe County guidance which is available on the county website (<http://www.thewasteauthority.com>) and any contractors can call the Monroe County Municipal Waste Management Authority at (570) 643-6100. The contractor shall provide a disposal certificate or landfill weight slip to the COR for all solid waste disposed of during the performance of this contract. The COR will then provide copies of the disposal certificate or landfill weight slips with the quarterly consolidated C&D report from The Installation Planning and Maintenance Division to EB.

-- End of Section --

SECTION 22 15 19.19 20

NONLUBRICATED ROTARY SCREW AIR COMPRESSORS (100 HP AND LARGER)
05/11

PART 1 GENERAL

1.1 SAFETY

Construct all components of the unit in accordance with the requirements of OSHA 29 CFR 1910.219. Requirements include shaft coupling guards as specified by manufacturer with manufacturer standard covering or aluminum sheet of all surfaces at 125 degrees F and higher within a height of 7 feet from floor level, and use of electrical safety devices. Thermal insulation, furnished by equipment manufacturer, must conform to ASTM C553, Type I (flexible resilient), Class B-5 (up to 400 degrees F), 2 pcf nominal. Cement insulation to surface with MIL-A-3316, Class 2, adhesive and fasten with 16-gage wire bands at maximum 16 inches on center spacing. Cover insulation with ASTM B209 sheet aluminum jacket. However, insulation is not required for hot piping inside sound enclosure.

1.2 EQUIPMENT ARRANGEMENT

Arrangement selected must maintain 3 foot clearance for access passage and 4 foot clearance for personnel to operate equipment. There are substantial physical and connection point differences among the several air compressors which comply with this specification. The Contractor must be responsible for selecting equipment and submitting arrangement drawings covering required changes for approval by the Contracting Officer. Changes from the equipment arrangement shown on the contract drawings must be performed by the Contractor at no additional cost to the Government.

1.2.1 Air Compressor System

Include wiring diagrams of the air compressor with all accessories. The minimum acceptable scale is 1/4 inch to one foot.

1.3 ELECTRICAL REQUIREMENTS

Comply with the requirements of NEC.

1.4 SUPERVISION

The Contractor must obtain the services of a qualified engineer or technician from the compressor manufacturer to supervise installation, start-up, and testing of the compressor. After satisfactory installation of the equipment, the engineer or technician must provide a signed certification that the equipment is installed in accordance with the manufacturer's recommendations.

1.5 DEFINITIONS

Conform to API Std 619 and the following:

Compressor power is shaft power at shaft coupling, including all losses and connected appurtenances.

1.6 INSULATION

Thermal and acoustical insulation must have flame spread rating not higher than 75, and smoke developed rating not higher than 150 when tested in accordance with [ASTM E84](#).

1.7 POSTED OPERATING INSTRUCTIONS

Provide for air compressor. Include start-up and shutdown sequence instructions.

PART 2 PRODUCTS

2.1 MATERIALS AND EQUIPMENT

Materials and equipment complete with accessories must be selected by the Contractor for performance compatibility.

2.2 AIR COMPRESSOR

The air compressors must be packaged, positive displacement rotary screw compressors capable of delivering oil-free air. No lubricant must be used within the compression chamber. Include air compressor, electric motor driver, coolers, lubrication system, and regulation and control systems mounted on a common base frame, and completely enclosed for noise control.

2.2.1 Manufacturer's Certifications

The manufacturer must certify that the air compressors proposed are of the same design, construction, size, and of equal or not more than 10 percent smaller in capacity as compressors which have been in satisfactory continuous service for at least 2 years at not less than two locations. Furnish the name of the owner, the address of the installation, and the name of a person at the installation who can be contacted for verification. The manufacturer must also certify that the factory is equipped to perform all required factory tests.

2.2.2 Guaranteed Performance

- a. Net compressed air output (All seal losses must be considered internal and not included in the net output) (plus or minus 2 percent): [1000 SCFM](#)
- b. Output pressure immediately downstream of aftercooler (minus zero plus 4 percent): [125 psig](#)
- c. Output air maximum temperature downstream of aftercooler: [100 degrees F](#)
- d. Inlet air pressure at first stage: [0 psig](#)
- e. Inlet air temperature at first stage: [ambient](#) degrees F
- f. Inlet air filtration efficiency: 99.9 percent of 0.5 micron size
- g. Barometric pressure: [0 psig](#)
- h. Relative humidity: [up to 80](#) percent

- l. Maximum compressor power required. (Plus or minus 4 percent): 200 hp
- m. Unloaded compressor power (maximum): 20% hp
- n. Maximum sound levels one meter horizontal from compressor and 1.5 meters above floor as measured per ISO 2151 Test Code for the Measurement of Sound from Pneumatic Equipment: 80 dBA, 90dB for any octave band.

2.2.3 Additional Performance Requirements

2.2.3.1 Air Quality

Air at compressor intake will be considered breathing air quality conforming to CGA G-7.1, Type I, Grade D or better. Air compressors must introduce no material, gases, or particles, or chemically alter any materials that will adversely affect or reduce the quality of the air passing through the unit.

2.2.3.2 Ambient and Inlet Conditions Operating Ranges

Allowing for rational engineering performance adjustments due to variations in ambient and inlet conditions, the compressor must be designed, equipped, and furnished to be fully operational without abnormal wear throughout the entire range between and including the limits of the winter and summer design conditions specified.

a. Summer Design Conditions:

Inlet Air: 86 degrees F dry bulb and 70.6 degrees F wet bulb temperatures, 50 percent relative humidity, Ambient Compressor Room Temperature: 95 degrees F, Barometric Pressure: 0 psig

b. Winter Design Conditions:

Inlet Air: 3.5 degrees F dry bulb 50 percent relative humidity, Inlet Ambient Compressor Room Temperature: 75 degrees F, Barometric Pressure: 0 psig.

2.2.3.3 Critical Speeds

Actual critical speeds must not encroach upon operating speed ranges at specified loads ranges. Rotors must be of a stiff shaft construction with the first actual rotor bending critical speed at least 120 percent of the maximum operating speed.

2.2.3.4 Vibration and Balance

Major parts of rotating elements, such as rotors, gears, and similar items must be individually dynamically balanced. During the factory and site tests of the assembled machine at operating speed, the double amplitude of vibration in any plane measured on the shaft adjacent and relative to a radial bearing must not exceed the limits of API Std 619, paragraph 2.7.2.5. For shafts which are not accessible, the manufacturer must submit a testing procedure to the Contracting Officer for approval.

2.2.4 Electrical Service Conditions

2.2.4.1 Air Compressor Drive Motor

480 volts, 3 phase, 3 wire, 60 hertz electrical service.

2.2.4.2 Accessory electrical Service

See Table I.

TABLE I - COMPRESSOR ACCESSORY ELECTRICAL SERVICE SCHEDULE			
Item	Voltage	Phase	Frequency
Control Power and Motors under 1/2 hp	120	1	60 Hz
Accessory Power	460	3	60 Hz

2.2.5 Compressor Controls

Provide a complete load regulation and control system with the compressor. Provide additional electrical, electro-pneumatic, or solid state electronic controls for other specified control and monitor functions. All electrical controls must conform to NEMA ICS 2 as selected by the compressor manufacturer. Control system enclosure must conform to NEMA ICS 6. Controls must be suitable for individual operation of the compressor or parallel operation with one or more other compressors.

2.2.5.1 Compressor Start-Up

The compressor must start unloaded. The manual starting circuit for the compressor must have interlocks to prevent the compressor drive motor from starting until pre-lubrication pump (if provided), oil pressure, and cooling water pump water flow have been established to the required values for safe operation as determined by the compressor manufacturer.

2.2.5.2 Load Regulation

The compressor must operate continuously at constant speed after being started. Provide means to load and unload the compressor automatically at preset minimum and maximum pressure settings. Minimum pressure must be 100 psig, and maximum pressure must be 125 psig. Unloading must be accomplished by a combination of closing the inlet valve and bypassing or venting the outlet of the compressor; however, input power at fully unloaded operation must not exceed 20 percent of full load input. Bypassed air must be cooled by the bypass cooler and if returned to the inlet of the first stage through an internal loop and must be limited to the minimum flow required to maintain compressor cooling. Air vented to the atmosphere when unloading need not be cooled.

2.2.5.3 Monitor and Safety Controls

Provide supplementary electric, electro-pneumatic, or solid state electronic controls to provide alarm and shutdown requirements, plus interlocks with accessories. Requirements are as follows:

- a. Shutdown requirements must cause the controlled compressor to shut

down, energize alarms, and light labeled red lights.

- b. Alarm only requirements must not cause the controlled compressor to shut down, but must sound the same alarms and light labeled amber lights.
- c. Light only requirements must not cause the controlled compressor to shut down, but must light labeled amber lights.
- d. The individual monitor and safety controls must be as shown on Table 2.

TABLE 2 - MONITOR AND SAFETY CONTROL SCHEDULE			
<u>Item</u>	<u>Light and Shutdown</u>	<u>Indicating Alarm</u>	<u>Light Only</u>
1. High Discharge Air Temperature 275 degrees F	Yes	Yes	-
2. High Intercooler Discharge Water Temperature, Each Intercooler	No	Yes	-
3. High Aftercooler Discharge Water Temperature	No	Yes	-
4. High Cooling Water Supply Temperature	No	Yes	-
5. High Lube Oil Temperature	Yes	Yes	-
6. Low Lube Oil Pressure	Yes	Yes	-
7. Low Cooling Water Flow	No	Yes	-
8. Low Oil Reservoir Level	No	Yes	-
9. High Condensate Level Intercooler (wired to one light)	No	No	Yes
10. High Bleed-Off Air Pressure	Yes	Yes	-
11. High Motor Stator Temperature	Yes	Yes	-
12. High Condensate Level Aftercooler	No	No	Yes
13. High Inlet Pressure Drop Across Inlet Air Filters (combined, 3 stage)	No	Yes	-
14. High CO Level	Yes	Yes	-

2.2.5.4 Monitoring Instruments

Provide the following monitoring instruments in addition to the monitor and safety controls. Pressure gages must conform to ASME B40.100, 4 1/2 inch diameter, red marking pointer, single bourdon tube, brass case, black

enamel finish. Provide pressure gages with a pressure snubber and a stainless steel barstock needle isolation valve. Thermometers must be extended stainless steel sheathed bimetallic stem, 3 1/2 inch dial, and separable 4 inch stainless steel wells. Temperature measurements at inaccessible locations must be made with remote reading thermometers conforming to MIL-T-19646, Class C separable well of Type 304 stainless steel. Select pressure and temperature gage ranges to give a normal operating reading near the midpoint of the scale range.

- a. Oil cooler outlet temperature gages for oil.
- b. Oil cooler inlet temperature gages for water.
- c. Lubrication oil bearing supply pressure gage.
- d. Compressor seal air pressure gage (if applicable).
- e. Inlet air filter differential pressure gage with 8, zero, 8 inch water gage. Provide selector valve, tubing, and tap to measure static gage pressure downstream of each filter stage.
- f. Total running time readout.
- g. Cooling water supply to compressor pressure gage.
- h. Cooling water return from compressor pressure gage.
- i. Interstage air pressure gages for each interstage.
- j. Compressed air pressure downstream of aftercooler pressure gage.
- k. Compressed air temperature downstream of aftercooler temperature gage.
- l. Compressed air temperature at discharge of each stage of compression before cooling temperature gages.
- m. Interstage air temperature after intercooler of each stage temperature gages.
- n. Compressor inlet air temperature gage.
- o. Cooling water to compressor temperature gage.
- p. Cooling water outlet temperature at each outlet of each intercooler, aftercooler, and bypass air cooler temperature gages.

2.2.5.5 Gages on Schematics

Certain pressure and temperature gages are designed on schematic flow diagrams in the drawings. Where a monitor gage satisfies the required location on a schematic, no additional gage needs to be furnished.

2.2.5.6 Control Schematics

The drawings show a generalized overall control system for compressor, auxiliaries, remote panel transmitting and receiving, and remote panel. The system is shown using relay symbology. Contractor and equipment suppliers may use standard panel features to accomplish the total requirements using other methods of signal, solid state devices, or

revised lamping. All wiring diagrams and required devices must be approved by the Contracting Officer prior to installation.

2.2.6 Compressor Design Features

The compressor must be a multistage, oil-free rotary screw compressor, with a minimum of two compressor stages, flanged to an integral speed increaser. Each stage must be driven from a common bull gear to ensure optimum speed and efficiency. An intercooler must be provided between stages and aftercooler must be provided after the final stage of compression. Silencers, lubricating system, cooling system, control system, and driver must be mounted as part of the package. Provide a common base frame for the compressor system and driver. Provide a sound enclosure over the compressor and driver. Equipment must be designed for economical and rapid maintenance. Casing components, bearing housings, and other major parts must be shouldered, dowelled, or designed with other provisions to facilitate accurate alignment or reassembly. Shaft seals and bearings must be accessible for inspection or replacement with a minimum of disassembly; however, compressors with compression elements (air end) provided as a factory-assembled not repairable in the field may be approved by the Contracting Officer if determined to be in the interest of the Government.

2.2.6.1 Casings

Casings must be cast iron, ductile iron, cast steel, or fabricated steel. Casing stresses must be within the limits allowed by [ASME BPVC SEC VIII D1](#). Casings, supports, and baseplates must be designed and fabricated to preclude excessive and injurious distortion from temperatures, pressures, and forces encountered in service conditions. Provide jackscrews, lifting lugs, eyebolts, guide dowels, and casing alignment dowels to facilitate disassembly and reassembly. When using jackscrews for parting contacting faces, relieve one of the faces by counterboring or recessing to prevent marring the face, which result in leaking or improper fit. Provide lifting lugs or eyebolts for removable portions of the casings. Flanged casing connections for external piping must conform to [ASME B16.1](#) or [ASME B16.5](#). Threaded connections for external piping must conform to [ASME B1.20.1](#). Air compression portion of the casing must be one-piece and must be provided with integral coolant passages and a large inlet port. Gear cases must be enclosed, accessible, force lubricated, and designed with seals and slingers to keep oil out of air system.

2.2.6.2 Shafts

Shafts must be of forged or rolled alloy steel and must have a machined finish throughout their entire length. All rotating components must be positively secured to shafts by approved mechanical means or interference shrink fits.

2.2.6.3 Rotors

Rotors must be steel, and of one-piece construction, with an asymmetric profile to minimize leakage losses, and ensure high efficiency. Rotors must be treated for corrosion resistance. If rotors are welded to the shaft, the assembly must be stress relieved and heat treated for proper strength. Rotors must be dynamically balanced to ensure vibration-free operation.

2.2.6.4 Gears

Gears must be of alloy steel, [ANSI/AGMA 2009](#) and [AGMA 2011](#) Quality Number 12 or better for both bull and pinion gears. Gears must be hardened to 275 Brinell for bull gear and 320 Brinell for pinion, unless otherwise approved. Gears must be ground to the required contours, checked for proper contact during assembly at the factory, and must not require a break-in period in the field for proper operation. All gears must be pressure lubricated.

Timing gears must be provided on the rotor shafts to maintain the rotors in correct relative position. The compressor design must allow the timing gears to absorb no more than 10 percent of the total input power at full load.

2.2.6.5 Seals

Separate air and oil shaft seals must be provided to confine air in the casing and prevent contamination of the air stream by lubricating oil. Shaft seals must be the restrictive ring type. The seal rings must be stainless steel, brass, or carbon, and retainers must be made of stainless steel. Provide an air space vented to the atmosphere between the air and oil seals. Seals must be suitable for all operating conditions including suction throttling, start-up, and shutdown.

2.2.6.6 Thrust Bearings

Thrust bearings must be anti-friction ball or roller type or hydrodynamic (fluid film) type. Anti-friction bearings must have an L-10 life of 80,000 hours in accordance with [ABMA 9](#) or [ABMA 11](#). Axial rotor thrusts due to air compression must be absorbed by main thrust bearings or transferred to auxiliary thrust bearings by a load balancing arrangement. Hydrodynamic thrust bearings must be Kingsbury type or other approved type and must be adequate to accommodate all operating conditions. Speed increaser bull gear thrust bearings must be sized for equal thrust in both directions and must be adequate for any axial loads transmitted through the driver coupling.

2.2.6.7 Radial Bearings

Radial bearings must be anti-friction roller or ball type or hydrodynamic type. Anti-friction bearings must have an L-10 life of 40,000 hours in accordance with [ABMA 9](#) or [ABMA 11](#). Hydrodynamic bearings must be precision bored sleeve or pad type, designed for easy replacement by a split design or axially removable arrangement. High speed hydrodynamic pinion bearings must be anti-oil whip, tilting pad type. Hydrodynamic bearing design must provide low vibration and sufficient damping at rated speed and all operating modes, including rated capacity and unloading down to 20 percent of unloaded power.

2.2.6.8 Speed Increaser

The speed increaser must be an integral part of the compressor unit and must include the main drive shaft and bull gear. The main drive shaft must be supported through anti-friction bearings.

2.2.6.9 Intercoolers, Aftercooler, Bypass Cooler, and Oil Coolers

Intercoolers, aftercooler, bypass cooler, and oil cooler must include

ASTM B111/B111M admiralty brass or other corrosion resistant tubes in ASTM B171/B171M admiralty or steel tube sheets and baffles for optimum cooling and fouling resistance using fresh water. Provide an intercooler between stages of compression factory assembled on unit base with piping. The aftercooler must be mounted on the unit base. Intercoolers, aftercooler, bypass cooler, and oil cooler must be factory tested at 1.5 times operating pressure. External intercoolers and aftercooler must be constructed in accordance with ASME BPVC SEC VIII D1 requirements and be ASME code stamped for 150 psig working pressure. Intercoolers and aftercooler must be capable of one piece bundle removal. Intercoolers and aftercooler must be equipped with an integral or direct connected moisture separator with condensate trap assembly. Design intercoolers and aftercooler for 20 and 15 degrees F approach, respectively; however, the approach temperature used to size the coolers must be reduced if required to meet aftercooler maximum air outlet temperature specified. Nonstandard coolers must be provided if required to meet the aftercooler maximum air outlet temperature requirement. All coolers must be of counter-flow design, with a fouling factor of 0.002 for both sides of the coolers.

2.2.6.10 Lubrication System

Include an integral sump, positive displacement pump, oil cooler, and twin filter\strainer (readily replaceable cartridges while operating). Provide a prelube lubrication oil pump for start-up and standby for hydrodynamic bearings or if required by the compressor design. System must be factory assembled and tested. Lubricating oil must conform to recommendations of the compressor manufacturer. Spray lubricate drive gear, anti-friction bearings, and timing gear in each stage. Pressure lubricate hydrodynamic bearings. Provide the oil sump with a level indicator and drain and fill connections.

- a. Prelubrication pump, if required, or motor-driven main lubrication pump must be sized by air compressor manufacturer for the requirements of the system, but must meet the following requirements. Pump must be positive displacement gear pump separately mounted with motor on a common base plate with drip lip and drain.
 - (1) Performance: Pump must have separate safety valve bypass set at 25 psi above peak expected pressure.
 - (2) Materials must be hardened steel gears and shaft, cast iron case, bronze bearings, mechanical seal.
 - (3) Flexible coupling with shaft guard must be provided, except that these items are not required for a close-coupled pump.
 - (4) Motor must be NEMA MG 1, Design A or B, Class B insulation, of open drip-proof type. Furnish combination type starter for motor.
- b. Lube Oil Heater: Provide thermostatically controlled electric heater in lubrication oil sump of sufficient capacity to heat up and maintain manufacturer's recommended oil temperature when unit is cold at 32 degrees F ambient. Provide low oil level indicator with light for protection of heater.

2.2.7 Electric Motors

2.2.7.1 Main Electric Drive Motor

The main drive motor for each compressor must be [an induction,] [or] [a synchronous] motor, 200 horsepower (hp), with a continuous service factor of 1.0. Size the motor so that the name plate hp rating is not exceeded under the entire range of operating conditions specified. Efficiency and losses must be determined in accordance with IEEE 112. Unless otherwise specified horizontal polyphase squirrel cage motors rated one to 125 horsepower must be tested by dynamometer Method B as described in Section 6.4 of IEEE 112. Motor efficiency must be calculated using Form B of IEEE 112 calculation procedures. Polyphase motors larger than 125 horsepower must be tested in accordance with IEEE 112 with stray load loss determined by direct measurement or indirect measurement (test loss minus conventional loss). The efficiency must be identified on the motor nameplate by the caption NEMA Nominal efficiency or NEMA Nom eff. Electrical service will be as specified. Motor must be designed for reduced voltage starting [at [50] [65] [80] percent of full voltage], allowing for characteristics of the connected load, and must start without undervoltage tripping. Provide resistance temperature detectors (RTD) attached to or imbedded in motor winding for control system. The motor must meet the requirements of NEMA MG 1 with Class F insulation. Provide space heaters for protection of windings during motor shutdowns.

2.2.7.2 Accessory and Related Equipment Motors

Motors less than 1/2 hp must be single-phase induction motors and must conform to NEMA MG 1. Motors 1/2 through 5 hp must be three-phase induction motors and must conform to NEMA MG 1. Single-phase and three-phase motors must have bimetallic disk thermostats attached to or imbedded in the motor winding. Motors must have NEMA MG 1 Class B insulation.

2.2.8 Control Panel

Control unit panel must conform to NEMA ICS 6, floor or frame mounted, factory designed, and assembled, and must be provided complete. The panel must be fabricated of formed stretcher leveled sheet steel, reinforced, and assembled into a rigid unit. Gasketed access doors must be provided as required. Panel must be factory finish painted. The panel must meet NEMA 12, requirements.

- a. Panel must contain electric and safety control work required, including either alarm annunciator or individual labeled pilot lights arranged in a group. Panel must contain alarm device with light and silencing. Generalized arrangement in accordance with drawings.
- b. Panel must contain start and stop buttons (the latter with lockout feature), discharge air pressure gage, control test switch and lights, reset button, green unit running light, and control selector switch.
- c. Oil pressure gages must be mounted separately from panel.

2.2.9 Accessories

Required accessories include:

2.2.9.1 Control Valves

Pneumatically or hydraulically controlled valves on suction inlet of compressor and on bypass or vent line.

2.2.9.2 Intake Devices

Compressor air inlet must be piped to the outside of the building and consist of the following:

- a. Intake weather hood with rain hood and bird screen. Material must be galvanized steel or aluminum alloy, minimum 20 gage.
- b. Intake pipe, [ASTM A36/A36M](#) steel galvanized, 12 gage or Schedule 5 minimum, from intake weather hood to filter housing flange, welded construction.
- c. [Filter housing](#) by filter manufacturer to include filter frames, access door(s). Material for housing must be [0.065 inch](#) thick, Class 5000 aluminum alloy. Unit must be rigid and free from distress with all seams sealed.
- d. Intake Pipe from Filter Enclosure to Compressor: Aluminum alloy [ASTM B209](#), Alclad alloy 5052-H32 or equivalent, minimum 10 gage, flanged, welded with 5XXX welding rod using TIG method and including expansion bellows.

2.2.9.3 Outlet Connectors

Compressor air outlet flexible connection of stainless steel bellows with braided steel cover jacket, with stainless steel liner sleeve, 18-inch (457-mm) nominal length bellows, flanged ends, Class 150. If air bypass connects separately to the compressor from the outlet line, provide a second flexible connection of stainless steel bellows with braided jacket for the bypass.

2.2.10 Inlet Air Filters

Provide a three-stage filter system, complete with mounting racks (horizontal flow), interstage seals, and replaceable filters. Filter unit must be provided complete including enclosure or housing, and frames. Enclosure must be Class 5000 aluminum alloy with inlet and outlet flanges. Construction must be welded or, where welding is not practical, close riveted and caulked, weathertight, with access doors for filter replacement and cleaning. Access doors must be reinforced, fully gasketed with continuous flexible neoprene gaskets, corrosion-resistant continuous hinges and quarter-turn latches to ensure tightness. All internal ferrous surfaces, including galvanized, must receive a factory-applied epoxy prime and finish coat for corrosion resistance. Filters must consist of three separate stages and sized to fit the available space.

2.2.10.1 First-Stage

First-stage filter must be flat, [2 inch](#) thickness, replaceable media, and rated for the required air quantity at [500 FPM](#) nominal face velocity, friction clean [0.25 inch water](#) gage, efficiency 98 percent of 15 microns and 90 percent of 5 microns.

2.2.10.2 Second-Stage

Second-stage filter must be deep pleated type, 9 inches nominal depth and rated for the required air quantity at 350 FPM nominal face velocity, friction clean 0.20 inch water gage, efficiency 98 percent of 5 microns and 90 percent of 3 microns.

2.2.10.3 Third-Stage

Third-stage filter must be deep pleated type 12 inches minimum depth and rated for the required air quantity at 350 FPM nominal face velocity, friction clean 0.30 inch water gage, efficiency 99.9 percent of 0.5 micron.

2.2.10.4 Filter Ratings

Filter media must be rated and listed UL Class 2. Filter efficiencies must be based on National Bureau of Standards (NBS) type discoloration gravimetric test method using atmospheric dust.

2.2.11 Bypass or Vent Line Silencer

A bypass or vent line silencer must be furnished with each compressor as selected by compressor manufacturer for sufficient noise attenuation to meet OSHA sound level criteria, but not greater than 84 dBA measured at an elevation of 5 feet, and 10 feet horizontally from silencer.

2.2.12 Sound Attenuating Enclosure

The compressor package, including the driver motor, must be contained within a noise reducing enclosure. Design of the enclosure must be such as to limit noise transmission to 84 dBA or less at a distance of one meter from the compressor in any direction.

2.2.12.1 Enclosure Frame

The enclosure frame must be designed to support the weight of the sound suppression panels and easily demountable. Connections to the base frame must be designed to allow the enclosure frame to be detached and lifted away without damage to the connections, enclosure frame or base frame, and to allow accessibility and replacement of any component.

2.2.12.2 Panels

The panels must be of rigid construction to allow repeated access without damage or distortion. Sound absorbing material must be mineral fiber, treated to preclude shedding of fibers. Other approved insulation may be used except that polyurethane foam must not be permitted. Top panels must be secured to the enclosure frame with quick disconnect fittings and fabricated to allow easy hand removal for maintenance. End and side panels must be hinged or lift out with positive closure latches. Panels must be designed to allow the maximum access area when opened. Provide acoustic seals as required. Controls and instrumentation mounted on the panels must have flexible connections for panel opening and disconnects for enclosure removal. Disconnects must be of the male-female plug type. Panels must split around all piping connections to allow enclosure removal without detaching piping. Controls must be visible and operable from outside the enclosure.

2.2.12.3 Ventilation

Fan(s) and sound baffled ventilation grilles must be provided as part of the enclosure. Ventilation must be sufficient to limit interior temperature to that required for cooling the motor.

2.2.13 Isolating Pad

If specifically recommended by the compressor manufacturer, each compressor steel or iron base frame must be mounted on a neoprene waffle or rib type isolator pad which extends uniformly and continuously along the base mounting surface. The neoprene material must be of bridge bearing pad quality neoprene and must be formulated for 40 durometer hardness. The maximum bearing pressure on the isolating pad must be 50 psi. The pads must be composed of two layers or 5/16 inch neoprene bonded to and sandwiching 16 gage galvanized steel. Compressor bolt down through the pad must be accomplished using 1/4 inch thick neoprene impregnated duck washers. Neoprene bushings are not acceptable.

2.3 AIR FLOW RATE AND PRESSURE RECORDER AND MEASUREMENT

Provide a complete flow and pressure measurement and recording package. Provide orifice flanges with pressure taps, square edged stainless steel paddle orifice plate. The orifice plate must be concentric type, of 0.125 inch thickness and must meet ASME Standards. Orifice must be sized for 40 inch water column differential at a full scale flow rate of 1000 SCFM at compressor based on 120 psig upstream pressure. Static gage pressure measurement device of the recorder must have a range of zero to 200 psig. Provide copper interconnecting tubing between the pressure taps and the recorder as part of this measurement and recording package. Provide a two-pen recorder for the measurement station. Pens must record pressure (0 to 200 psig range) and air flow (0 to 1000 SCFM). Recorder must be electric drive and housed in dust-tight steel cabinet. Charts must be 12 inch diameter with evenly divided graduations. Drive must be 7 day circle. Provide continuous flow integration of a 7 digit counter type. Pens must be supplied with long-life cartridges and capillary supply. Chart case must be internally illuminated. Access to charts must be through front access window door. Calibrated overall accuracy of the recorded measurements must be within plus or minus 1.0 percent of full scale. Furnish a supply of 400 charts with the recorder.

2.4 CARBON MONOXIDE MONITOR

The carbon monoxide (CO) monitor unit must be of the pressure type with attached sampling system. The unit must be solid state type operation, 2 to 50 ppm range, CO indicating, with provisions for milliamp signal to remote recorder, adjustable set point, and normally open/normally closed contacts for remote signal. Power must be 120 volt, single phase, 60 hertz with power cord and plug. Response time normally 2 minutes per sample/purge. Unit must be mounted in a gasketed enclosure with face gage indicating CO readings.

2.4.1 Sampling System

Sampling system must include shutoff valve filter/regulator, pressure gage, manual drainer, and line humidifier set at 50 percent. Draw sample from compressor discharge.

2.4.2 Test System

Test system must include calibration gas (20 ppm CO) cylinder test gas (200 ppm CO) cylinder, and calibration connectors with quick disconnect.

2.5 SOURCE QUALITY CONTROL

2.5.1 Factory Test Procedures

The completely assembled air compressor package including the actual contract drive motor, intercooler, lubrication system, and control panel must be subjected to [performance tests](#) and [sound level and run-in tests](#). Unit must comply with guarantee requirements applying engineering adjustments to guarantee conditions. Test must be certified by the manufacturer. Test may be run on the manufacturer's test stand using driver for this contract. Tests must be in accordance with [ASME PTC 9](#) format. Full-range performance tests must indicate performance at maximum rated flow, rating point, and unloaded conditions. Motor performance conditions must be reported, including motor efficiency and losses, motor power factor, motor service factor, motor temperature rise, motor noise and balance, and motor torque at full load, locked rotor, pull up, and break down. Include intercoolers, aftercoolers, and lubrication and control systems performance. Completed unit must be factory tested with sound meters in accordance with [ISO 2151](#). Location must be one horizontal meter from unit at 1.5 meters above the floor. Test must include readings at each octave band midpoint and the "A" scale, and must not exceed 84 dBA and 90 decibels at any octave band. Results of test must be included in the factory test report on the [ISO 2151](#) format. Factory test data may be corrected to the levels of an equivalent background noise level of 60 dBA showing calculations for reference use.

2.5.2 Supervision of Testing

System and components testing must be conducted or supervised by either a designated authorized and factory trained representative of the compressor manufacturer supplying the unit or a registered Mechanical Engineer experienced in such work.

2.5.3 System Test

Testing of system must conform to requirements outlined and must be witnessed by the Contracting Officer.

2.5.4 Approval of Testing Procedure

Proposed testing procedure must be approved by the Contracting Officer and the individual in charge of testing prior to conducting tests.

2.5.5 Certification of Performance Tests

The test supervisor must certify performance by test to be in compliance with specifications.

PART 3 EXECUTION

3.1 INSTALLATION

The Contractor must install the air compressors and accessories in accordance with manufacturer's recommendations and as indicated on the

drawings. All equipment must be installed plumb and level and anchored to structure, matching holes provided. Install the compressor under the direct supervision of an authorized representative of the manufacturer.

3.2 GENERAL REQUIREMENTS FOR INSTALLING AIR COMPRESSORS

Air compressors with contract motor and accessories must be factory assembled, run in, and tested complete before shipment to job site. The Contractor is advised that there are limitations to door opening sizes and available crane lifting capacity. Crane unit is specified to permit single lifts of complete compressor under special approval only. Should the unit require disassembly for installation, reassembly must be under the direct supervision of the compressor manufacturer's authorized representative. Complete unit must be mounted on a rigid single or equivalent mechanically joined steel or iron base. Submit installation sequence plans to the Contracting Officer for approval prior to installation. Any building materials removed to accomplish installation must be reinstalled if undamaged, by removal procedures; or if damaged, must be replaced with new materials to match original configuration.

3.2.1 Prompt Installation

The Contractor is advised that any compressor received must be installed and placed in operation promptly to prevent time deterioration when not installed. Should the Contractor sustain a delay exceeding 90 days prior to actual installation, the Contracting Officer must have the option of requiring breakdown and reassembly to inspect and clean prior to placing in operation. This work must be at no additional cost to the Government.

3.2.2 Start-Up Services

The Contractor must furnish the services of a compressor manufacturer's authorized representative to supervise prestart checkout, initial start-up, performance testing, and operator instruction. Time available must be as required to properly start up but not less than 3 consecutive days for the compressor.

3.3 FIELD QUALITY CONTROL

3.3.1 Field Test Procedures

Complete field performance testing of the total system must be performed by the Contractor and witnessed by the Contracting Officer. [Air compressor system test](#) must be conducted by either a compressor manufacturer's factory trained and authorized representative approved by the Contracting Officer or a qualified registered Mechanical Engineer. Tests may be run on individual components or on the system as a whole at Contractor option. Field tests require use of the actual compressor drive motor. Test must include operation at rated capacity for not less than 4 hours.

3.3.1.1 Air Compressor Performance Tests

Complete performance test must be run at maximum load, rated load, at point of unload but prior to unload, and unloaded condition. Data must be recorded listing:

- a. Air flow, inlet pressure and temperature, humidity; discharge pressure and temperature.

- b. Intercooler water flows, temperatures, and pressures.
- c. Aftercooler water flow, temperatures, and pressures.
- d. Bypass cooler water flow, temperatures, and pressures.
- e. Lube oil cooling water flow, temperatures, and pressures.
- f. Lube oil flow, pressures, and temperature.
- g. Cooling water pump flow, pressures, and motor amperage.
- h. Closed circuit cooler air flow, water and air temperatures, water pressure, and motor amperage.
- i. Electrical load in volts and amperes for compressor motor (loaded and unloaded), prelube oil pump motor, and compressor auxiliaries.
- j. Intake filter pressure differential (clean).
- k. Start-up sequence, alarm signals and automatic system shutdown.
- l. Test compressor intake and discharge for conformance to CGA G-7.1. Compressor discharge must show no increase in contaminants.

3.3.1.2 Instrumentation Test

The Contractor may use instrumentation provided in the contract and instrumentation provided by the Contractor to conduct the test. The testing procedure and instrumentation must be submitted to the Contracting Officer for approval prior to conducting tests. The format of ASME PTC 9 is required. It is intended that a full field test be performed. However, in lieu of precise instrumentation, the Contractor may use certified cooling water pump curves and closed circuit cooler. Shutdown signals must be caused by throttling selected fluids. Test data, such as air intake temperature and humidity, must be mathematically corrected to performance test requirement levels.

3.3.1.3 Sound Level Tests

Sound level tests must be conducted concurrently. Broad Band "A" scale readings and Octave Band readings must be taken and recorded at the same positions as on the factory testing. Maximum permissible level must be 84 decibels one horizontal meter from the compressor and 1.5 meters above the floor, with unit in operation and all other significant equipment not required for test within the same building bay shutdown at the same location previously described. A background noise correction to 60 decibels is permissible.

3.3.1.4 Operational Deficiencies

Any operational deficiencies noted in the tests must be promptly corrected and affected portions of the test rerun.

3.3.1.5 Testing Tolerances

A tolerance of plus or minus 2 percent on flow, plus or minus 4 percent on power, or plus or minus 5 percent on any other variable for each item of

equipment or fluid with all others conforming is permissible on field test results when compared to factory test data and to guarantee performance data except that compressor air flow, discharge pressure, and motor power must be met.

3.3.2 Approval of Testing Procedure

Proposed testing procedure must be approved by the Contracting Officer and the individual in charge of testing prior to conducting tests.

3.4 TRAINING OF GOVERNMENT PERSONNEL

During start-up and field testing, train Government station personnel in the operation and maintenance of compressor, associated equipment, and all control and safety devices. Training must not commence until equipment is operational and station personnel are in attendance. At least one day of classroom training and one day of field training must be furnished for each designated Government personnel. When factory training is required by the compressor manufacturer for proper maintenance and overhaul of the compressors, such training must be furnished by the compressor manufacturer at no additional cost to the Government. The Government will bear the cost of travel and living expenses for Government personnel as necessary for the factory training.

-- End of Section --